HU Collection

Test of Mathematics for University Admission

Mock Set D Paper 2

Mathematical Reasoning

Time allowed:	**75 minutes**
Questions:	20 multiple choice
Calculators:	not permitted
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Date written:	24 September 2026

Name: _____

Date sat: _____

Score: _____ / 20

Instructions

Answer all 20 questions. Each question has exactly one correct option and is worth one mark. No marks are deducted for wrong answers. Throughout, "must be true" means true in every case allowed by the information given, and a counterexample must satisfy the hypothesis while failing the conclusion.

Q	1	2	3	4	5	6	7	8	9	10
Ans										
Q	11	12	13	14	15	16	17	18	19	20
Ans										

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Question 1

Which of the following is the negation of the statement

"In every class, some student has solved every problem."

A In every class, no student has solved every problem.
B In some class, some student has not solved any problem.
C In some class, every student has failed to solve at least one problem.
D In every class, every student has failed to solve at least one problem.
E In some class, no student has solved any problem.
F In some class, some student has failed to solve at least one problem.

Question 2

Consider the statement: *if $f'(a) = 0$, then the curve $y = f(x)$ has a turning point at $x = a$.* Which of the following is a counterexample to the statement?

A $f(x) = x^2$, $a = 0$
B $f(x) = x^3$, $a = 0$
C $f(x) = x^4$, $a = 0$
D $f(x) = x^3 - 3x$, $a = 1$
E $f(x) = x$, $a = 0$
F $f(x) = x^4 - 2x^2$, $a = 0$

Question 3

For a positive integer n, the condition "n is divisible by 4" is _____ for "$n^2 + 3n$ is divisible by 4".

A necessary but not sufficient
B sufficient but not necessary
C necessary and sufficient
D neither necessary nor sufficient

Question 4

Each of six cards has a whole number on one side and a shape on the other. The visible faces are

4, 7, triangle, circle, 10, square.

The rule is: *if a card has an even number on one side, then it has a triangle on the other side.* Which cards must be turned over to check whether the rule holds for all six cards?

A 4 and 10 only
B 4, 10 and the triangle
C 4, 7 and 10
D 4, 10, the circle and the square
E the circle and the square only
F all six cards

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Question 5

Which of the following statements are true for every integer n?

I $n^3 - n$ is divisible by 6
II $n^5 - n$ is divisible by 30
III $n^2 + n + 1$ is odd
IV $n^2 - 1$ is divisible by 8

A I only
B I and III only
C II and III only
D I, II and IV only
E I, II and III only
F I, II, III and IV

Question 6

Real numbers a and b satisfy $a > b$. Which of the following must be true?

I $a^2 > b^2$
II $a^3 > b^3$
III $\frac{1}{a} < \frac{1}{b}$, provided a and b are non-zero
IV $2^a > 2^b$
V $a|a| > b|b|$

A II only
B II and IV only
C II, IV and V only
D I, II and IV only
E II, III, IV and V
F all five

Question 7

Which of the following are true for every x with $0° < x < 90°$?

I $\sin x + \cos x > 1$
II $\sin^2 x < \sin x$
III $\tan x > 1$
IV $\cos x < \frac{1}{\cos x}$

A I and II only
B II and IV only
C I and IV only
D I, III and IV only
E I, II, III and IV
F I, II and IV only

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Question 8

Five integers have the property that the sum of any three of them is even. Which of the following must be true?

I All five integers are even.
II The sum of all five integers is even.
III The sum of any two of the integers is even.
IV Each of the integers is divisible by 4.

A I only
B I and II only
C II and III only
D I, II and III only
E I, II, III and IV
F none of them

Question 9

Consider the statement: *if f and g are both decreasing on the real line, then the product fg is increasing on the real line.* How many of the following pairs are counterexamples to this statement?

I $f(x) = e^{-x}$, $g(x) = e^{-x}$
II $f(x) = -e^{x}$, $g(x) = -e^{x}$
III $f(x) = -x$, $g(x) = -x$
IV $f(x) = -x^3$, $g(x) = -e^{x}$

A 1
B 2
C 3
D 4

Question 10

A triangle has sides $a \leq b \leq c$, with the angle C opposite c. Which of the following conditions is sufficient for the triangle to be acute-angled?

I $C < 90°$
II $a^2 + b^2 > c^2$
III $a = b = c$
IV $b^2 + c^2 > a^2$

A I, II and III only
B I and II only
C III only
D II and IV only
E I, II, III and IV
F II only

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Question 11

An integer N with $1 \leq N \leq 30$ is the subject of five statements:

A: N is prime. B: N is even. C: N is a perfect square. D: $N > 10$. E: N is a multiple of 3.

Exactly three of the five statements are true, and statement C is one of them. Find N.

A 4
B 9
C 12
D 16
E 25
F there is no such N

Question 12

A student proves: *if* $a > 0$, $b > 0$ *and* $a + b = 2$, *then* $ab \leq 1$.

(I) Suppose, for contradiction, that $ab > 1$.
(II) Then $b > \dfrac{1}{a}$.
(III) So $a + b > a + \dfrac{1}{a}$.
(IV) But $a + \dfrac{1}{a} \geq 2$ for every $a > 0$.
(V) Hence $a + b > 2$, which contradicts $a + b = 2$.

Which of the following is correct?

A The proof is valid.
B Line II is wrong: dividing by a may reverse the inequality.
C Line IV is false for some $a > 0$.
D Line V is wrong: from lines III and IV we may only conclude $a + b \geq 2$.
E The argument proves the converse of the claim.
F The claim itself is false.

Question 13

A student adapts the standard proof that $\sqrt{2}$ is irrational to "prove" that $\sqrt{4}$ is irrational.

(I) Suppose $\sqrt{4} = \dfrac{p}{q}$, where p and q are positive integers with no common factor.
(II) Then $p^2 = 4q^2$, so p^2 is even and hence p is even.
(III) Write $p = 2k$. Then $4k^2 = 4q^2$, so $k^2 = q^2$.
(IV) Hence q^2 is even, so q is even.
(V) So p and q are both even, a contradiction. Hence $\sqrt{4}$ is irrational.

In which line does the first error occur?

A I
B II
C III
D IV
E V
F there is no error

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Question 14

The cubic $f(x) = x^3 + bx^2 + cx + d$ satisfies $f(0) < 0$, $f(1) > 0$ and $f(2) < 0$. Which of the following must be true?

I f has three distinct real roots.
II $b < -3$
III f has a local minimum between $x = 1$ and $x = 2$.
IV $d < 0$

A I and IV only
B I, II and IV only
C I, II, III and IV
D II and IV only
E I and III only
F IV only

Question 15

Which of the following are equal to $\int_{-1}^{1} f(x)\,dx$ for *every* quadratic function f?

I $\frac{1}{3}\left(f(-1) + 4f(0) + f(1)\right)$
II $f\left(-\frac{1}{\sqrt{3}}\right) + f\left(\frac{1}{\sqrt{3}}\right)$
III $2f(0)$
IV $f(-1) + f(1)$

A I only
B II only
C I and II only
D I and III only
E I, II and III only
F I, II, III and IV

Question 16

Each of the following statements is either true or false.

1. Exactly one of these three statements is true.
2. Statement 1 is false.
3. Statements 1 and 2 are both false.

Which statements are true?

A statement 1 only
B statement 2 only
C statement 3 only
D statements 2 and 3 only
E no assignment of truth values is consistent
F more than one assignment is consistent

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Question 17

A student solves $x^4 - 3x^3 + 4x^2 - 3x + 1 = 0$ over the real numbers.

(I) $x = 0$ is not a solution, so divide by x^2: $x^2 + \dfrac{1}{x^2} - 3\left(x + \dfrac{1}{x}\right) + 4 = 0$.

(II) With $t = x + \dfrac{1}{x}$, we have $x^2 + \dfrac{1}{x^2} = t^2 - 2$, so $t^2 - 3t + 2 = 0$ and $t = 1$ or $t = 2$.

(III) For every non-zero real x, $x + \dfrac{1}{x} \geq 2$, so $t = 1$ is impossible.

(IV) $t = 2$ gives $x^2 - 2x + 1 = 0$, so $x = 1$.

(V) Hence the only real solution is $x = 1$.

Which of the following is correct?

A The argument is correct.

B Line III is false, but the conclusion is correct.

C Line III is false, and the conclusion is wrong.

D Line II contains an algebraic error.

E Line IV is wrong: $x + \frac{1}{x} = 2$ has two real solutions.

F Line I is wrong, since $x = 0$ must be checked separately.

Question 18

The function $f(x) = x^3 + ax^2 + bx$ is called *increasing* if $f(x_1) < f(x_2)$ whenever $x_1 < x_2$. Which condition on a and b is necessary and sufficient for f to be increasing?

A $a^2 < 3b$

B $b > 0$

C $b \geq 0$

D $a^2 \leq b$

E $a^2 \leq 3b$

F $a = 0$ and $b \geq 0$

Question 19

Which of the following conditions on the real numbers b and c is necessary and sufficient for the equation $x^2 + bx + c = 0$ to have two distinct positive roots?

A $b < 0$ and $c > 0$

B $b^2 > 4c$ and $c > 0$

C $b < 0$ and $b^2 > 4c$

D $c > 0$ and $b < -2\sqrt{c}$

E $b < -2\sqrt{c}$

F $b < 0$, $c > 0$ and $b^2 \geq 4c$

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Question 20

The sequence $u_1, u_2, u_3, \ldots$ is defined by $u_1 = 1$ and $u_{n+1} = u_n + \dfrac{2}{u_n}$. Which of the following are true?

I $u_{n+1} > u_n$ for every n.
II $u_n^2 > 4n - 4$ for every $n \geq 2$.
III $u_{100} > 20$
IV $u_{100} > 30$

A I only
B I and II only
C I, II and IV only
D I and III only
E I, II and III only
F I, II, III and IV

END OF PAPER

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